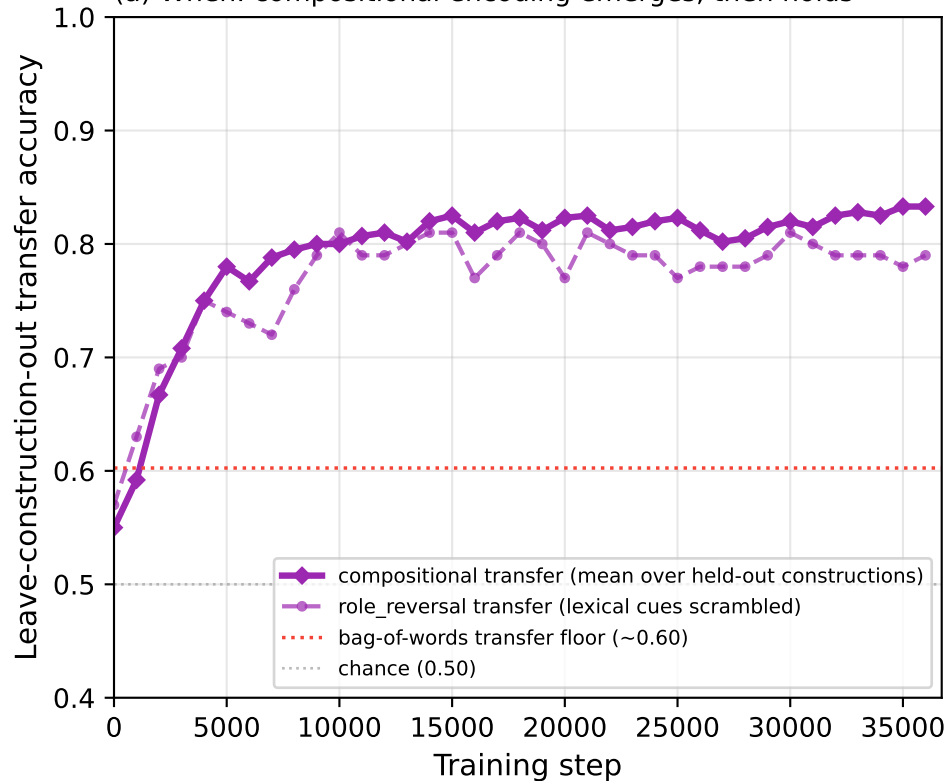


Compositional moral encoding emerges during pre-training (OLMo-2 1B early-training, 37/37 checkpoints)

(a) When: compositional encoding emerges, then holds



(b) Where: per-layer in-distribution probe accuracy

